

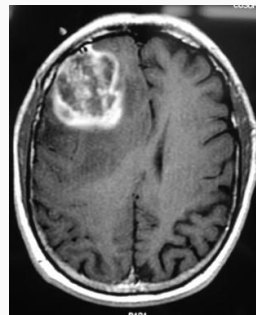
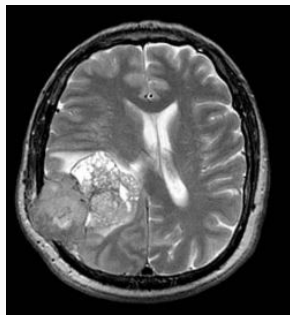
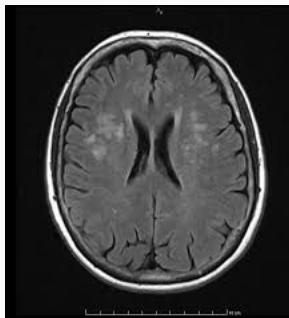
Neural Networks for Tumor Detection and Classification from MRI

Calvin Tyler



Refresher Summary

This project will focus on evaluating existing architectures for image classification and the criteria for selecting an appropriate model. After selection, a pretrained model will be modified and retrained to detect and classify tumors from MRI scans.



Purpose & Objectives

- Benefit to society
- Learn CNNs
 - Existing Models
 - Transfer Learning



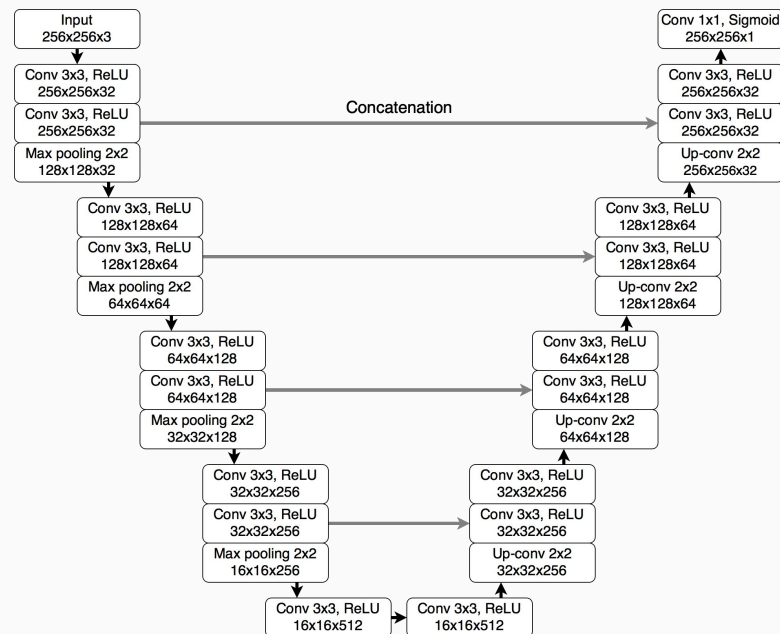
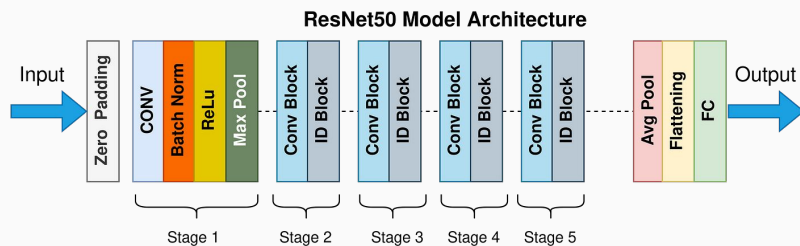
Models

U-Net - specifically designed for biomedical image segmentation

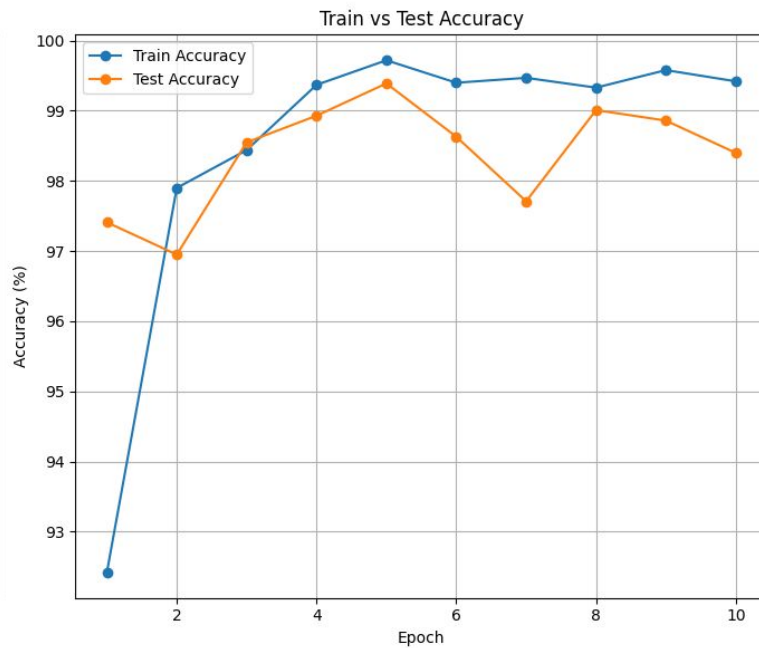
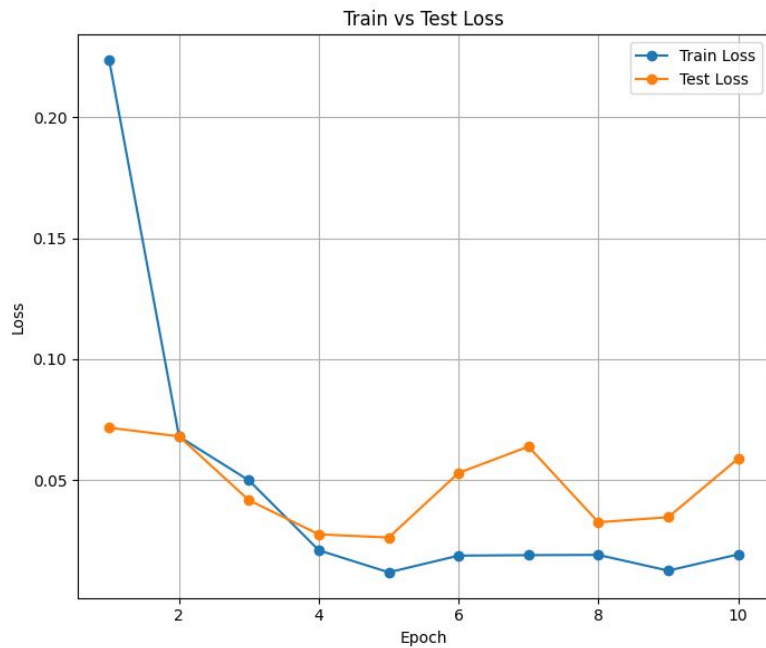
~~DenseNet - it performs well even with limited amounts of labeled data~~

ResNeXt - builds on the proven ResNet architecture

~~3D CNN (3D U Net or 3D ResNet) - volumetric data~~



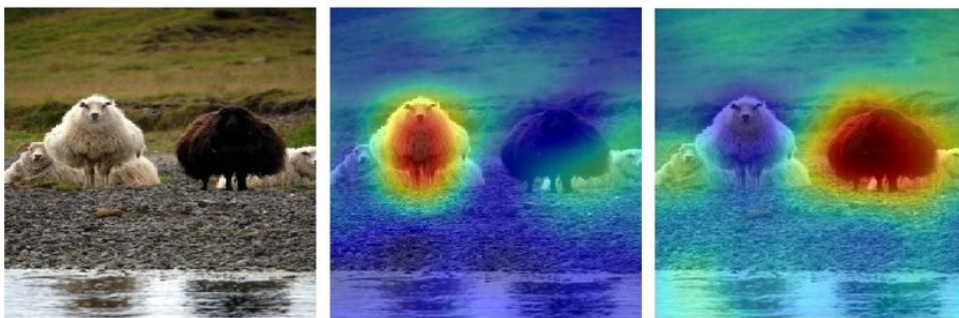
Results



Potential Further Exploration

- Compare multiple models
- Streamline inputs
- Create new architecture

Interpretability of the model



(a) Sheep - 26%, Cow - 17%

(b) Importance map of 'sheep'

(c) Importance map of 'cow'

Conclusion

- Learning lots
- Initial scope accomplished
- Future exploration explored
- Explainability goals added!

